

Adeesha - Fun Projects

Project 1: AI Music Generator

An AI-powered music composition tool that generates original melodies based on user input moods and styles.

Tech Stack: Python, TensorFlow, MIDI processing, Flask

Features:

- Mood-based music generation (happy, sad, energetic)
- Multiple genre support (jazz, classical, lo-fi)
- Real-time playback and MIDI export
- Web interface with interactive controls

Duration: 2 weeks

Status: Completed

GitHub: github.com/adeesha/ai-music-gen

Project 2: Retro Game AI Bot

A reinforcement learning agent that learns to play classic arcade games using Deep Q-Networks.

Tech Stack: Python, PyTorch, OpenAI Gym, Atari

Features:

- Self-learning agent using DQN algorithm
- Experience replay and target network
- Training visualization with TensorBoard
- Score tracking and model checkpoints

Results:

- Achieved human-level performance on Breakout
- Trained 10M frames in 4 hours on single GPU

Duration: 3 weeks

Status: Completed

Project 3: Smart Recipe Generator

An NLP-powered app that generates recipes based on available ingredients.

Tech Stack: Python, Hugging Face Transformers, FastAPI, React

Features:

- Ingredient-based recipe suggestions
- Dietary restriction filters
- Step-by-step cooking instructions
- Nutritional information calculator

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Duration: 1 week

Status: Completed